Hari Charan V

Electrical And Electronics Engineering

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Professional summary

I am an Electrical & Electronics Engineering student with solid expertise in the fields of renewable energy, power electronics, and intelligent control systems. I have hands-on experience working with MATLAB-Simulink, embedded systems, Internet of Things (IoT), and Printed Circuit Board (PCB) designing. I have worked on developing an artificial neural network-based maximum power point tracking controller for solar PV.

Skills

- Simulation & Design Tools: MATLAB/Simulink, LTspice, Keil Micro-Vision
- Programming Languages: Python, C, Java
- PCB Design (Ki-cad, Easy-EDA)
- Hybrid Systems
- Technical Skills
- Microcontrollers
- Embedded Systems (Microcontrollers, Microprocessors – Sensor Interfacing)
- Neural networks
- Internet of Things (IOT)
- Team collaboration
- Circuit Design

Experience

Intern, 05/2025 - 07/2025
Center for Smart Grid, VIT Chennai – Chennai, India
Guide: Dr. Sithartan
Worked on the design and performance evaluation of a Hybrid Power Generation System integrating solar PV, wind energy, and fuel cell sources.
My primary contribution focused on developing an Artificial Neural Network (ANN)-based Maximum Power Point Tracking (MPPT) controller for the solar PV subsystem to enhance power extraction efficiency.
Key responsibilities and achievements:

- Modeled a solar PV system and boost converter using MATLAB/Simulink based on the single-diode model
- Generated a comprehensive dataset by varying irradiance and temperature conditions for ANN training
- Designed and trained a feed-forward ANN (2 inputs, 15 hidden neurons, 1 output)

- Implemented the ANN-based MPPT controller to predict optimal duty cycle and improve system performance
- Achieved faster dynamic response and reduced steady-state oscillations compared to conventional MPPT techniques
- Explored the application of AI for system optimization and fault detection in hybrid energy systems

This experience strengthened my understanding of renewable energy systems, power electronics, and the application of machine learning in real-time control systems.

EDUCATION

B.Tech: Electrical And Electronics Engineering, Expected in 04/2027

Vellore Institute of Technology - Chennai

LANGUAGES

English

Bilingual or Proficient (C2)

Tamil

Bilingual or Proficient (C2)

Hindi

Bilingual or Proficient (C2)

Kannada

Advanced (C1)

ACTIVITIES

Activities & Leadership Active member of the Entrepreneurship Cell (E-Cell), contributing to organizing and managing college events. Served as the Auctioneer for an IPL-themed event, demonstrating leadership, communication, and event coordination skills. Member of the college Finance Club, participating in finance-related discussions and activities. Activities & Leadership Active member of the Entrepreneurship Cell (E-Cell), contributing to organizing and managing college events. Served as the Auctioneer for an IPL-themed event, demonstrating leadership, communication, and event coordination skills. Member of the college Finance Club, participating in finance-related discussions and activities.

Projects and Research:

- **Temperature Based Fan Speed Control with LED Indicators**
  Developed an automatic fan speed control system using Arduino Nano and DHT22 sensor. Implemented PWM-based regulation with LED indicators for different temperature ranges.
- **Optimization of PID Controller Parameters for Enhanced Motor Control**
  Optimized PID controller gains (Kp, Ki, Kd) using the Grasshopper Optimization Algorithm (GOA) to achieve faster motor response, reduced overshoot, and improved stability in MATLAB/Simulink.
- **DC-DC Buck Boost Converter Design and PCB Implementation**
  Designed and simulated an inverting buck-boost converter in MATLAB, analyzed voltage gain characteristics, and created a PCB schematic and layout using EasyEDA.

- **Adaptive Headlight Intensity Control System using Arduino (RC Car Implementation)** Developed an Arduino-based adaptive headlight system using LDR to automatically adjust brightness based on ambient light, demonstrated on an RC car for real-time glare reduction and improved safety.
- **AI-ML Driven State Estimation for Smart Grid Applications** Implemented an ANN-based power system state estimation model on the IEEE 14-bus system in MATLAB/MATPOWER, improving accuracy and robustness over traditional WLS methods under varying load scenarios.
- **Real-Time Power Monitoring and Power Factor Correction System:** Designed and implemented a real-time power monitoring and power factor correction system using a PZEM module, relay-based capacitor switching, and display interface to measure electrical parameters and enhance power factor.
- **SeizureGuard: Multi-Sensor Wearable Bio-Telemetry PCB:** Designed and routed a compact, low-noise PCB for health monitoring, featuring optimized I2C bus architecture for motion and temperature tracking combined with heartbeat and analog signal processing for skin conductance analysis.